

Microprocessor, microcontroller, and embedded system

Course code: CSE 3523

Credit Hour: 3

Md. Tohidul Islam Khan

Dept. of CSE, IIUC

masumkhan1802118@gmail.com

International Islamic University Chittagong(IIUC)

Instruction Set

ADDITION

```
MOV AX, 30F0H
MOV BX, 2000H
ADD [BX], AL
```

Draw memory and write flags change

DS Memory Map

FF	2000
3C	2001
0F	2002
FE	2003

$$\begin{array}{r} 1111\ 1111\ (\text{FFH} = -1\ \text{signed}) \\ +\ 1111\ 0000\ (\text{F0H} = -16\ \text{signed}) \\ \hline 1\ 1110\ 1111\ (\text{1EFH} - \text{carry bit}) \end{array}$$

EFH = 1110 1111

1. Carry Flag (CF):

- ✓ Carry out the Most Significant Bit (bit 7)
- ✓ There's a carry (1) beyond the 8th bit
- ✓ **CF = 1**

2. Zero Flag (ZF):

- ✓ EFH ≠ 00H
- ✓ Check if result (lower 8 bits) is zero
- ✓ **ZF = 0** ✓ (Correct)

ADDITION

3. Sign Flag (SF):

- ✓ Check Most Significant Bit of result (bit 7)
- ✓ EFH = 1110 1111 → MSB (bit 7) = 1
- ✓ **SF = 1** ✓ (Correct)

4. Parity Flag (PF):

- ✓ Count 1's in lower byte (EFH = 1110 1111)
- ✓ 1's: positions 7,6,5,3,2,1,0 = 7 ones
- ✓ 7 is odd → **PF = 0**

5. Auxiliary Carry Flag (AF):

- ✓ Check carry from bit 3 to bit 4
- ✓ Lower nibble: FH (1111) + 0H (0000) = FH (1111)
- ✓ No carry from bit 3 → **AF = 0**

6. Overflow Flag (OF):

Signed Interpretation:

- ✓ FFH as signed 8-bit = **-1**
- ✓ F0H as signed 8-bit = **-16**
- ✓ Expected: -1 + (-16) = **-17**

When does signed overflow occur?

- ✓ **Positive + Positive = Negative** (overflow)
- ✓ **Negative + Negative = Positive** (overflow)
- ✓ **Mixed signs** = never overflow

Here: **Negative + Negative = Negative** ✓ (no overflow)

CF=1, ZF=0, SF=1, OF=0, PF=0, AF=0

ADDITION

SHL (Shift Left) and **SAL (Shift Arithmetic Left)** are **the same operation**, but why are **SHR** and **SAR** different?

- ✓ **SHL** $\equiv$ **SAL** (identical operations)
- ✓ **SHR** $\neq$ **SAR** (different fill behavior)
- ✓ **SHR** for unsigned numbers
- ✓ **SAR** for signed numbers (preserves sign)